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SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' GRADE Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - NOV'2025

COMMON TO B.E – CSE & M.TECH - CSE

20CS259 - FOUNDATIONS OF BLOCKCHAIN TECHNOLOGY

Answer ALL questions

Duration: 2 Hrs 30 min

Maximum: 70 Marks

PART - A

(5x 1 = 5 Marks)

1. **CO1** Identify the primary purpose of Nakamoto consensus?
a) Increase bandwidth b) Achieve consensus
c) Enable faster mining d) Encrypt transactions
2. **CO2** Find the hashing algorithm which is predominantly used in Bitcoin?
a) MD5 b) SHA-1 c) SHA-256 d) SHA-512
3. **CO3** Label the following platforms uses Solidity language?
a) Hyperledger b) Ethereum c) Corda d) Bitcoin
4. **CO2** In Proof of Stake, block validators are selected based on?
a) Computational power b) Energy spent c) Stake held d) Transaction fee
5. **CO1** Enumerate the structure used in Bitcoin to verify transactions efficiently?
a) Binary Tree b) Hash Tree c) Merkle Tree d) AVL Tree

PART - B

(5x 3 = 15 Marks)

6. **CO1** Explain the use of Proof of Work (PoW) mechanism in Bitcoin.
7. **CO2** Describe the hardness of mining in Bitcoin.
8. **CO2** Differentiate between Hyperledger Sawtooth and Ethereum.
9. **CO2** Explain the functionality of Ethereum Virtual Machine (EVM).
10. **CO3** List the major attacks possible on smart contracts.

PART - C

(5 x 10 = 50 Marks)

Q.No: 11 Compulsory Question

Q.No: 12 to 17 Answer any FOUR Questions

Compulsory Question:

11. **CO1** i) A consortium of universities is planning to implement a blockchain system to verify academic certificates securely and instantly. Analyze how the Consensus Problem impacts such multi-party blockchain environments. (5)
- CO1** ii) Explain the working and significance of Nakamoto Consensus in ensuring trust in decentralized certificate validation. (5)
12. **CO2** i) Describe the structure and advantages of GARAY and RLA models for blockchain. (5)
- CO2** ii) Explain the Proof of Stake (PoS) based chains with suitable examples. (5)
13. **CO1** i) Apply cryptographic basics (hashing and signature schemes) to ensure transaction security in cryptocurrencies. (5)
- CO1** ii) Demonstrate transaction verification in Bitcoin with a practical example involving a sample transaction. (5)
14. **CO2** i) Summarize various types of wallets used in Ethereum. (5)
- CO2** ii) Develop a simple HelloWorld smart contract in Solidity. (5)
15. **CO5** i) Summarize the concept of Zero Knowledge Proofs and their relevance in Blockchain privacy. (5)
- CO3** ii) Discuss SNARK protocols and their applications. (5)
16. **CO2** i) Explain Hyperledger Fabric and Sawtooth architecture and their use cases. (5)
- CO2** ii) Describe the architecture and features of Corda blockchain. (5)

17. **CO3** A blockchain-based electronic voting system is proposed for a university student election. Each student will be registered with a unique account through Meta Mask Wallet. Votes will be recorded securely on a private blockchain created using Ganache. The voting functionality, including candidate registration and vote casting, is managed by a Solidity smart contract deployed on the local blockchain. (10)
- Design a basic smart contract structure in Solidity to manage candidate registration, voting process, and result declaration.
 - Analyze the advantages and possible challenges of using Ganache and Meta Mask for implementing such a voting system in a real-world university election scenario.
